

Relational Analysis (2)

CSE552 Program Analysis — Lecture 17

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Octagon Domain¹

- ▶ $Var = \{x_i \mid i \in I\}$
- ▶ $State =$
 $\{\perp\} \cup \left\{ \bigwedge_{j \in J} c_{j,1}x_{i_{j,1}} + c_{j,2}x_{i_{j,2}} \leq b_j \mid c_{j,1}, c_{j,2} \in \{-1, 0, 1\} \right\}$

¹The octagon abstract domain (Miné, 2006)

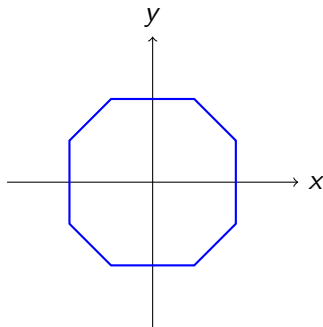
Octagon Domain — Properties

- ▶ Only two forms of inequalities are allowed
 - ▶ $\pm x \pm y \leq c$
 - ▶ $\pm x \leq c$
- ▶ Subsumes interval domain
 - ▶ $x : [l, h] \iff x \leq h \wedge -x \leq -l$

Intuition

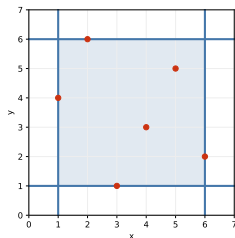
- Possible values of two variables are represented by an octagon (8 edges)

- $x \geq c_1$
- $x \leq c_2$
- $y \geq c_3$
- $y \leq c_4$
- $y \geq x + c_5$
- $y \leq x + c_6$
- $y \geq -x + c_7$
- $y \leq -x + c_8$

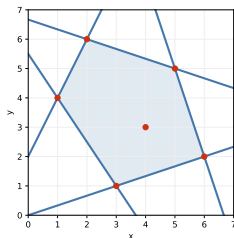


Comparison

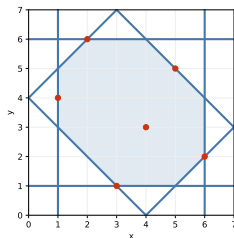
- ▶ $(x, y) = (1, 4), (2, 6), (3, 1), (4, 3), (5, 5), (6, 2)$
- ▶ Interval: $x = [1, 6], y = [1, 6]$
- ▶ Linear inequalities (polyhedra): $3y \geq x, y \leq -3x + 20, 3y \leq -x + 20, y \leq 2x + 2, 2y \geq -3x + 11$
- ▶ Octagon: $x \geq 1, x \leq 6, y \geq 1, y \leq 6, y \geq x - 4, y \leq x + 4, y \geq -x + 4, y \leq -x + 10$



Interval



Polyhedra



Octagon

Use Case 1: Dynamically Sized Arrays

```
x = input();  
y = [0; x];  
if z <= x - 1 {  
    y[z] = 1;  
}
```

► $z - x \leq -1$

Use Case 2: Binary Analysis

```
x = sp - 4;  
*x = 0;  
  
y = sp - 4;  
z = *y;
```

- ▶ $x \geq sp - 4, x \leq sp - 4$
- ▶ $y \geq sp - 4, y \leq sp - 4$
- ▶ $x - y \geq 0, x - y \leq 0$

Difference-Bound Matrices

- ▶ Potential constraint: $x_j - x_i \leq c$
- ▶ A set of potential constraints can be represented by a $N \times N$ difference-bound matrix (DBM) m :

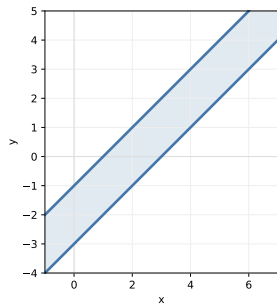
$$m_{ij} = \begin{cases} c & \text{if } x_j - x_i \leq c \\ \infty & \text{otherwise} \end{cases}$$

Potential Graphs

- ▶ For a DBM m , its potential graph is a weighted graph where
 - ▶ $Node = Var$
 - ▶ $Edge = \{(x_i, x_j) \mid m_{ij} < \infty\}$
 - ▶ $weight(x_i, x_j) = m_{ij}$
- ▶ A constraint $x_i - x_j \leq c$ is drawn as $x_j \xrightarrow{c} x_i$

DBM and Potential Graph — Example

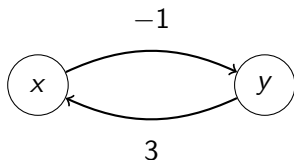
- ▶ $Var = \{x, y\}$
- ▶ $x - y \leq 3$
- ▶ $y - x \leq -1$



DBM:

$$m = \begin{pmatrix} \infty & -1 \\ 3 & \infty \end{pmatrix}$$

Potential graph:



Extending DBMs

- ▶ $Var = \{x_0, \dots, x_{n-1}\}$
- ▶ $V = \{x_0^+, x_0^-, \dots, x_{n-1}^+, x_{n-1}^-\}$
- ▶ Within a potential constraint,
 - ▶ a positive variable x_i^+ is interpreted as $+x_i$
 - ▶ a negative variable x_i^- is interpreted as $-x_i$

Notation

- ▶ Now, a DBM is a $2n \times 2n$ matrix m where
 - ▶ index $2i$ corresponds to x_i^+
 - ▶ index $2i + 1$ corresponds to x_i^-
- ▶ $\bar{i} = i \oplus 1$ where $\oplus$ is bitwise XOR
 - ▶ $\overline{2i} = 2i + 1$
 - ▶ $\overline{2i + 1} = 2i$
 - ▶ If i corresponds to x_a^+ then $\bar{i}$ corresponds to x_a^- and vice versa

Coherence

- ▶ m is coherent if and only if

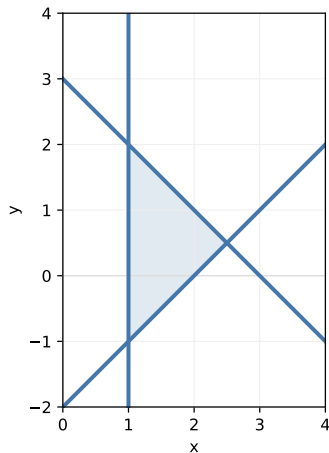
$$m_{ij} = m_{\overline{j}\overline{i}} \quad \text{for all } i, j$$

Representing Constraints

Over Var	Over V
$x - y \leq c$	$x^+ - y^+ \leq c, \quad y^- - x^- \leq c$
$x + y \leq c$	$x^+ - y^- \leq c, \quad y^+ - x^- \leq c$
$x \leq c$	$x^+ - x^- \leq 2c$

Extended DBM — Example (Constraints)

- ▶ $Var = \{x, y\}$
- ▶ $x - y \leq 2$
- ▶ $x + y \leq 3$
- ▶ $x \geq 1$



Extended DBM — Example (Matrix and Potential Graph)

► $V = \{x^+, x^-, y^+, y^-\}$

► $x^+ - y^+ \leq 2$

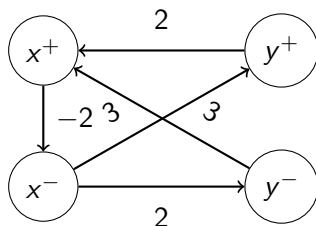
► $y^- - x^- \leq 2$

► $x^+ - y^- \leq 3$

► $y^+ - x^- \leq 3$

► $x^- - x^+ \leq -2$

$$m = \begin{pmatrix} \infty & -2 & \infty & \infty \\ \infty & \infty & 3 & 2 \\ 2 & \infty & \infty & \infty \\ 3 & \infty & \infty & \infty \end{pmatrix}$$

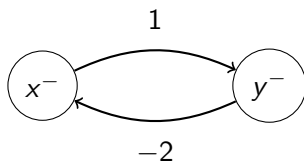
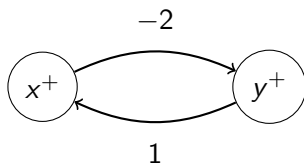


Emptiness Test

- ▶ Checks whether the state is actually $\perp$
- ▶ A state can be considered $\perp$ if its potential graph has a cycle with a strictly negative weight
- ▶ Uses Bellman–Ford algorithm

Emptiness Test — Example

- ▶ $x - y \leq 1$
- ▶ $x - y \geq 2$
- ▶ $V = \{x^+, x^-, y^+, y^-\}$
- ▶ $x^+ - y^+ \leq 1$
- ▶ $y^- - x^- \leq 1$
- ▶ $y^+ - x^+ \leq -2$
- ▶ $x^- - y^- \leq -2$



Normalization

- ▶ Two DBMs may represent constraint sets with the same solution set
- ▶ We can normalize DBMs so that a single solution set is represented by a unique DBM

Normalization — Example 1

$$\blacktriangleright x - y \leq 1$$

$$\blacktriangleright y - z \leq 2$$



$$\blacktriangleright x - y \leq 1$$

$$\blacktriangleright y - z \leq 2$$

$$\blacktriangleright x - z \leq 3$$

Normalization — Example 2

▶ $x \leq 1$

▶ $y \geq 2$



▶ $x \leq 1$

▶ $y \geq 2$

▶ $x - y \leq -1$

Order

- ▶ $s_1 \sqsubseteq s_2$ if and only if the solution set of s_1 is a subset of the solution set of s_2
- ▶ Where s_1 is represented by a DBM m and s_2 is represented by a DBM n , $s_1 \sqsubseteq s_2$ if and only if $m_{ij} \leq n_{ij}$ for all i, j
- ▶ Example: $\{x - y \leq 1\} \sqsubseteq \{x - y \leq 2\}$

Join and Meet

- ▶ Where s_1 is represented by a DBM m and s_2 is represented by a DBM n ,
 - ▶ $s_1 \sqcup s_2$ is represented by a DBM l where $l_{ij} = \max(m_{ij}, n_{ij})$
 - ▶ $s_1 \sqcap s_2$ is represented by a DBM l where $l_{ij} = \min(m_{ij}, n_{ij})$
- ▶ Example
 - ▶ $\{x - y \leq 1, z - w \leq 2\} \sqcup \{x - y \leq 2, z - w \leq 1\} = \{x - y \leq 2, z - w \leq 2\}$
 - ▶ $\{x - y \leq 1, z - w \leq 2\} \sqcap \{x - y \leq 2, z - w \leq 1\} = \{x - y \leq 1, z - w \leq 1\}$

Widening

- ▶ Same principle as for intervals
 - ▶ Keep stable inequalities
- ▶ Where s_1 is represented by a DBM m and s_2 is represented by a DBM n , $s_1 \nabla s_2$ is represented by a DBM l where
 - ▶ $l_{ij} = m_{ij}$ if $n_{ij} \leq m_{ij}$ and
 - ▶ $l_{ij} = \infty$ otherwise
- ▶ Example
 - ▶ $\{x - y \leq 1, z - w \leq 2\} \nabla \{x - y \leq 2, z - w \leq 2\} = \{x - y \leq \infty, z - w \leq 2\}$

Control-Sensitivity

- ▶ Condition: $x_a - x_b \leq c$

$$m'_{ij} = \begin{cases} \min(m_{ij}, c) & \text{if } (j, i) \in \{(2a, 2b), (2b+1, 2a+1)\} \\ m_{ij} & \text{otherwise} \end{cases}$$

- ▶ Condition: $x_a + x_b \leq c$

$$m'_{ij} = \begin{cases} \min(m_{ij}, c) & \text{if } (j, i) \in \{(2a, 2b+1), (2b, 2a+1)\} \\ m_{ij} & \text{otherwise} \end{cases}$$

- ▶ Condition: $x_a \leq c$

- ▶ Equivalent to $x_a + x_a \leq 2c$

Control-Sensitivity — Example

- ▶ State: $x - y \leq 2$
- ▶ Condition: $x - y \leq 3$
- ▶ Updated state: $x - y \leq 2$

- ▶ State: $x - y \leq 2$
- ▶ Condition: $x - y \leq 1$
- ▶ Updated state: $x - y \leq 1$

Assignments (Invertible)

- ▶ $x_a = x_a + c$
- ▶ $m'_{ij} = m_{ij} + (\alpha_{ij} + \beta_{ij})c$ where

$$\alpha_{ij} = \begin{cases} 1 & \text{if } j = 2a \\ -1 & \text{if } j = 2a + 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\beta_{ij} = \begin{cases} -1 & \text{if } i = 2a \\ 1 & \text{if } i = 2a + 1 \\ 0 & \text{otherwise} \end{cases}$$

Invertible Assignment — Example

- ▶ State: $x - y \leq 1, x \leq 2, z - x \leq 3$
- ▶ Assignment: $x = x + 1$
- ▶ Updated state: $x - y \leq 2, x \leq 3, z - x \leq 2$

Assignments (Non-Invertible)

► $x_a = x_b + c \ (a \neq b)$

$$m'_{ij} = \begin{cases} c & \text{if } (j, i) \in \{(2a, 2b), (2b+1, 2a+1)\} \\ -c & \text{if } (j, i) \in \{(2b, 2a), (2a+1, 2b+1)\} \\ m_{ij} & \text{if } i, j \notin \{2a, 2a+1\} \\ \infty & \text{otherwise} \end{cases}$$

Non-Invertible Assignment — Example

- ▶ State: $x - y \leq 1, z - w \leq 2$
- ▶ Assignment: $x = y + 5$
- ▶ Updated state: $x - y \leq 5, y - x \leq -5, z - w \leq 2$

Summary

- ▶ Octagons capture constraints of the form $\pm x \pm y \leq c$, strictly more expressive than intervals but cheaper than full polyhedra
- ▶ Difference-bound matrices (DBMs), extended with x_i^+ and x_i^- variables, encode octagonal states as $2n \times 2n$ matrices
- ▶ Potential graphs give a geometric view of DBMs; emptiness reduces to detecting a negative-weight cycle via Bellman–Ford
- ▶ Order, join, meet, and widening on octagons reduce to entrywise comparisons on DBMs
- ▶ Control-sensitivity and assignment transfer functions are defined index-wise on the DBM, with separate rules for invertible and non-invertible updates